BODY—BODY JOINT REINFORCEMENT MATERIAL (STRUCTURAL FOAM)—REPAIR PROCEDURES	Article No. 01-21-9

FORD: 2002 EXPLORER

MERCURY: 2002 MOUNTAINEER

ISSUE

The following document outlines repair procedures for collision-damaged vehicles that contain Structural Foam.

ACTION

Inspect collision-damaged sheet metal for structural foam and replace it following the repair procedure below.

BACKGROUND

Structural Foam

Structural Foam is a new technology material, which is being used in vehicles to reinforce joints, body panels and cavities. This material is not the same as Acoustical Foams, Sound Deadeners, and Hem Adhesives. Structural Foam's are significantly different from any other foam product currently on the market. There are two types of structural foam applications that are being used in vehicles today.

Carrier Application

The first application type is referred to as a carrier application. This application uses a thin layer of Structural Foam attached to a Steel or Nylon reinforcement. These parts are then put into the vehicle in locations such as A-pillars, D-pillars, Frame Rails, and Underbody Joints. When these parts go through the OEM E-coat baking process the Structural Foam expands and the result is a joint that is reinforced from the inside where traditional welding would not reach. The 2002 Explorer/Mountaineer Structural Foam applications are of this type.

Full-Fill Application

The second application type is referred to as a full-fill application. In this application, a two-part epoxy Structural Foam is pumped into selected locations in the body sheet metal or frame that require reinforcing. The result is a fill of cement-like material that adheres to the metal and creates a solid joint. The epoxy cures at room temperature and does not require high temperature ovens.

SERVICE PROCEDURE

NOTE

READ THIS REPAIR PROCEDURE COMPLETELY BEFORE PERFORMING ANY SERVICE ACTIONS.

Structural Foam is being used in the following seven locations on the Explorer and Mountaineer vehicles. The * designation represents application that have been discontinued after July 2001 production dates. The ** designation represents applications that have been discontinued after January 2002 production dates. Note: Right and Left as seated in vehicle.

- Lower A-pillar - Right and Left sides *
- Lower D-pillar - Right and Left sides **
- Underbody between the #1 and #2 body mounts - Right and Left sides
- Underbody between the #2 and #3 body mounts - Right and Left sides *
- Underbody #2 cross member cap - Right and Left sides *
- Underbody forward, in front of rear wheel well (forward waterfall) - Right and Left sides *
- Underbody rear, in front of rear wheel well (rear waterfall) - Right and Left sides *

Whether body parts are being replaced or repaired, you must do one of the following:

- If sheet metal is being replaced in any of these mentioned locations you will need to fully fill the cavities.
- If (repairing not replacing body sheet metal) the Structural Foam parts appear to be damaged you must full-fill each area.

This will encase reinforcement parts in place and eliminate rattles. The repair procedures should be done after sheet metal work has been completed and before final painting of the vehicle.

Copyright © 2001 Ford Motor Company

Cartridge Get Ready Procedure (X)

Use Motorcraft (TA-12) two-part epoxy to fill the cavities of the vehicle that need to be repaired. Epoxy can be applied using either a manual or pneumatic application gun. Foam flow can be significantly affected by temperature and will flow easily when it reaches a temperature range of 46°-74°C (115°F-165 °F). It is highly recommended that the Structural Foam cartridges be heated before use. Put the cartridges in a paint booth during a bake cycle or place the cartridges in a bucket of hot water for approximately 20 minutes. Remember the cartridge is hot so gloves may be required for handling. Once the foam is heated, place it in the application gun and dispense a small amount until the material is fully mixed inside the mixing nozzle. The material is now ready to be applied.

Lower A-Pillar - Right And Left Sides

This application will require 10 cartridges of foam.

1. Apply aluminum duct tape to cover over the small rectangular hole on the inside of the door jam (Figures 1 and 2).
2. Apply aluminum duct tape to cover the two 5mm trim attachment holes and the 25mm hole on the A-pillar inner panel, right side only (Figure 1).
3. Cut and insert a 100x50x75mm (4x2x3 inch) soft foam block into the forward most rectangular build up hole in the side sill inner panel (Figure 3).
4. Follow the foam Get Ready Procedure (X) to begin application.
5. Apply the foam into the lower A-pillar through the large rectangular hole located on the A-pillar inner panel (Figures 1 and 2).
6. Allow the foam to cure for 60 minutes and remove aluminum tape.

Lower D-Pillar - Right And Left Sides

This application will require 13 cartridges of foam.

1. Insert a 140x100x75mm (5.5x4x3 inch) soft foam block into the #4 body mount area. The front edge of the block should be outboard of the trim attachment. The block will prevent foam expansion into the body mount (Figure 4).
2. Apply aluminum duct tape to cover two 6.5 mm holes, the large rectangular (16mm by 27mm) hole and the 10 mm diameter hole that are inside the vehicle at the lower D-pillar (Figure 5).
3. Apply aluminum duct tape over the 35mm slotted hole, small oval hole, and square hole in the lower D-pillar just above the D-pillar bottom (stuffer plate) (Figure 6).
4. Apply aluminum duct tape to the flanges of the bottom of the D-pillar "stuffer plate" all four corners should be completely sealed. Apply aluminum tape to the two (slotted holes) and two 12mm diameter holes. (Figure 6).
5. Apply aluminum duct tape to the tab hole and 8 mm diameter holes in the rear of the vehicle (Figure 6).
6. Follow the foam Get Ready Procedure (X) to begin application.
7. Dispense the foam into the lower D-pillar through the 18.5mm hole located at the #4 body mount (Figure 4).
8. Allow the foam to cure for 60 minutes and remove aluminum tape.

Underbody Between The #1 And #2 Body Mount - Right And Left Sides

This application will require 4 cartridges of foam.

1. If removed, install the interior trim rubber grommet plug into the 25 mm hole in the floor pan trough (Figure 7).
2. If the vehicle is not equipped with running boards, apply aluminum duct tape over the running board fastener attachment holes (three sets) in the underbody frame beneath the area to be filled. If the vehicle is equipped with running boards, keep the J-clips and fasteners for the running boards in these holes and remove the fastener after 40 minutes of cure time.
3. Follow the foam Get Ready Procedure (X) to begin application.

4. Dispense 2 cartridges of the foam into each fastener hole for a total of 4 cartridges. If repairing a part that has carrier structural foam still in place, drill a 6mm (1/4 ″) hole between the two rivets and dispense all 4 cartridges into this hole (Figure 7).
5. Apply duct tape over the dispensing holes when application is completed.
6. Visually make sure the Structural Foam does not leak into the #2 body mount. Any Structural Foam leakage will need to be cleaned out.
7. Allow the foam to cure for 60 minutes and remove aluminum tape.

Underbody Between The #2 And #3 Body Mounts - Right And Left Sides

This application will require 7 cartridges of foam.

1. Before the rocker panel outer is installed, apply aluminum duct tape to the (oval fixture) hole in the inner side sill (Figure 8). If it is not possible to cover this hole, the repair will require an extra cartridge to obtain a full fill of foam.
2. Apply aluminum duct tape to cover the 34.5 mm square hole and 10 mm diameter round hole in the bottom of the floor pan substructure (ladder assembly) forward of the #3 cross member (Figure 8).
3. Apply aluminum duct tape to the 20 mm square and 12 mm diameter round running board fastener attachment holes rearward of the #3 cross member (Figure 8).
4. Drill a 6 mm (1/4″) hole centered between the two-fastener attachment holes in the floor pan just inboard of the trough (Figure 8).
5. Follow the foam Get Ready Procedure (X) to begin application.
6. Apply 7 cartridges of Structural Foam into the previously drilled 6 mm (1/4″) hole.
7. Allow the foam to cure for 60 minutes and remove aluminum tape

Underbody #2 Cross Member Cap - Right And Left Sides

This application will require 3 cartridges of foam.

1. Apply aluminum duct tape to the openings where the cross member meets the underbody longitudinal structural rails that attach the underbody cross rails at the outside edges of the floor pan on the underside of the vehicle (Figure 9).
2. Apply aluminum duct tape to the 30.5 mm square hole located on the underside of the #2 cross member (Figure 9).
3. Drill a 6 mm (1/4″) hole into the floor pan at the center of the fill cavity (Figure 9).
4. Follow the foam Get Ready Procedure (X) to begin application.
5. Dispense 3 cartridges of Structural Foam into the hole previously drilled in the cavity.
6. Apply duct tape over the fill hole when application is completed.
7. Allow the foam to cure for 60 minutes and remove aluminum tape.

Underbody Forward, In Front Of Rear Wheel Well (Forward Waterfall) - Right And Left Sides

This application will require 5 cartridges of foam.

1. Run the seat and the seat belt anchor bolts back into their respective holes (out board holes only). This will prevent the structural foam from clogging these holes. If foam fills the holes, the holes must be drilled out before these bolts can be reinserted. After the material has set for 40 minutes, the bolts can be removed for reattaching the seats and the seat belt anchor (Figure 10).
2. Apply aluminum duct tape over the inside wall of the #3 body mount (Figure 10).
3. Apply aluminum duct tape over the 25 mm diameter hole in the trough of the floor pan (Figure 10).
4. Apply aluminum duct tape over the 20mm and 40mm slotted holes in the underbody rail (Figure 10).

5. Follow the foam Get Ready Procedure (X) to begin application.
6. Dispense 2.5 cartridges of Structural Foam into each of the fastener holes for a total of 5 cartridges. If a repair is being made instead of a replacement and the carrier part is still in place drill a 6 mm (1/4 ″) hole between the two fastener holes for dispensing purposes (Figure 10).
7. Apply duct tape over the holes when the application is completed.
8. Allow the foam to cure for 60 minutes and remove aluminum tape.

Underbody Rear, In Front Of Rear Wheel Well (Rear Waterfall) - Right And Left Sides

This application will require 5 cartridges of foam.

1. Apply aluminum duct tape to cover the 19 mm diameter hole located on the outboard side of the underbody structural frame in the fender wheel well (Figure 11).
2. Insert a 100x50x75mm (4x2x3 inch) soft foam block into the 30 mm diameter hole in the bottom of the underbody structural frame, and then apply aluminum duct tape over the hole (Figure 11).
3. Follow the foam Get Ready Procedure (X) to begin application.
4. Dispense 2.5 Structural Foam cartridges into each fastener hole for a total of 5. If a repair is being made instead of a replacement and the carrier part is still in place drill a 6 mm (1/4 ″) hole between the two fastener holes for dispensing purposes (Figure 11).
5. Apply duct tape over the holes when application is completed.
6. Allow the foam to cure for 60 minutes and remove aluminum tape.

NOTE
STRUCTURAL FOAM KIT CONTAINS FOUR CARTRIDGES OF STRUCTURAL FOAM AND ONE FOAM BLOCK.

Vehicle Location	Structural Foam Cartridge Requirements	Foam Block Requirements
Lower A-Pillar	10	1
Lower D-Pillar	13	1
Underbody Between The #1 And #2 Body Mounts	4	N/R
Underbody Between The #2 And #3 Body Mounts	7	N/R
Underbody #2 Cross Member End Cap	3	N/R
Underbody Rear Wheel Well Forward	5	N/R
Underbody Rear Wheel Well Rear	5	1

PART NUMBER	PART NAME
TA-12	Motorcraft Structural Foam Kit
TA-10	Motorcraft Manual Application Gun
TA-11	Motorcraft Pneumatic Application Gun

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 111000, 702000

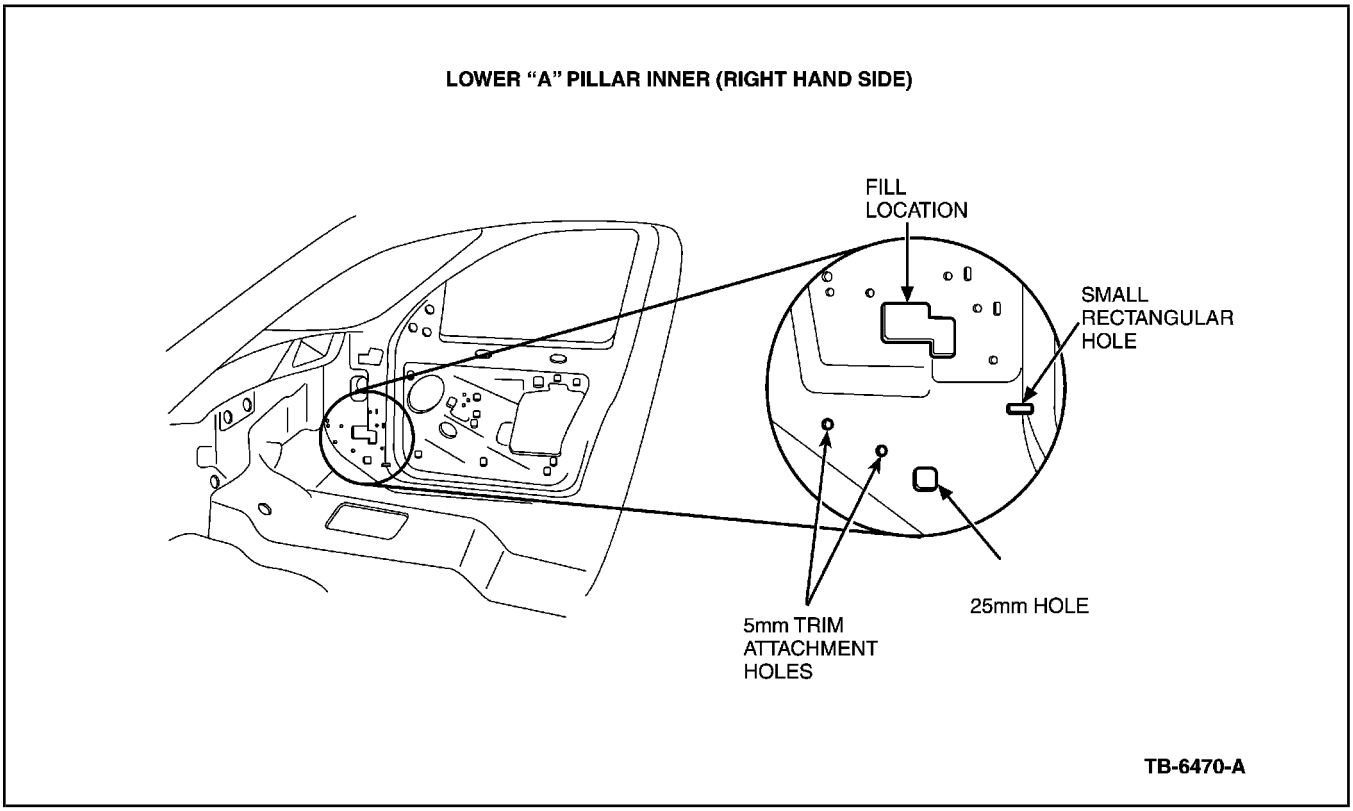


Figure 1 - Article 01-21-9

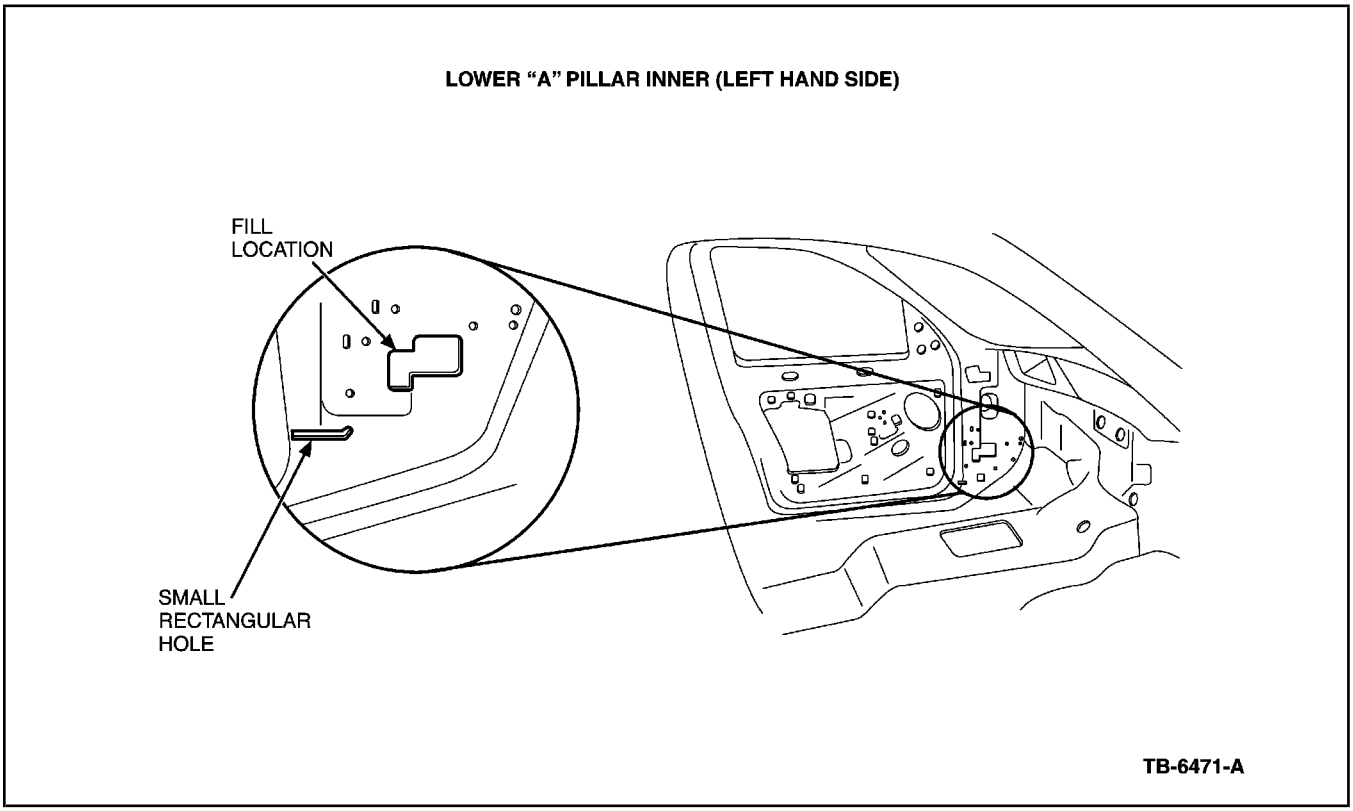


Figure 2 - Article 01-21-9

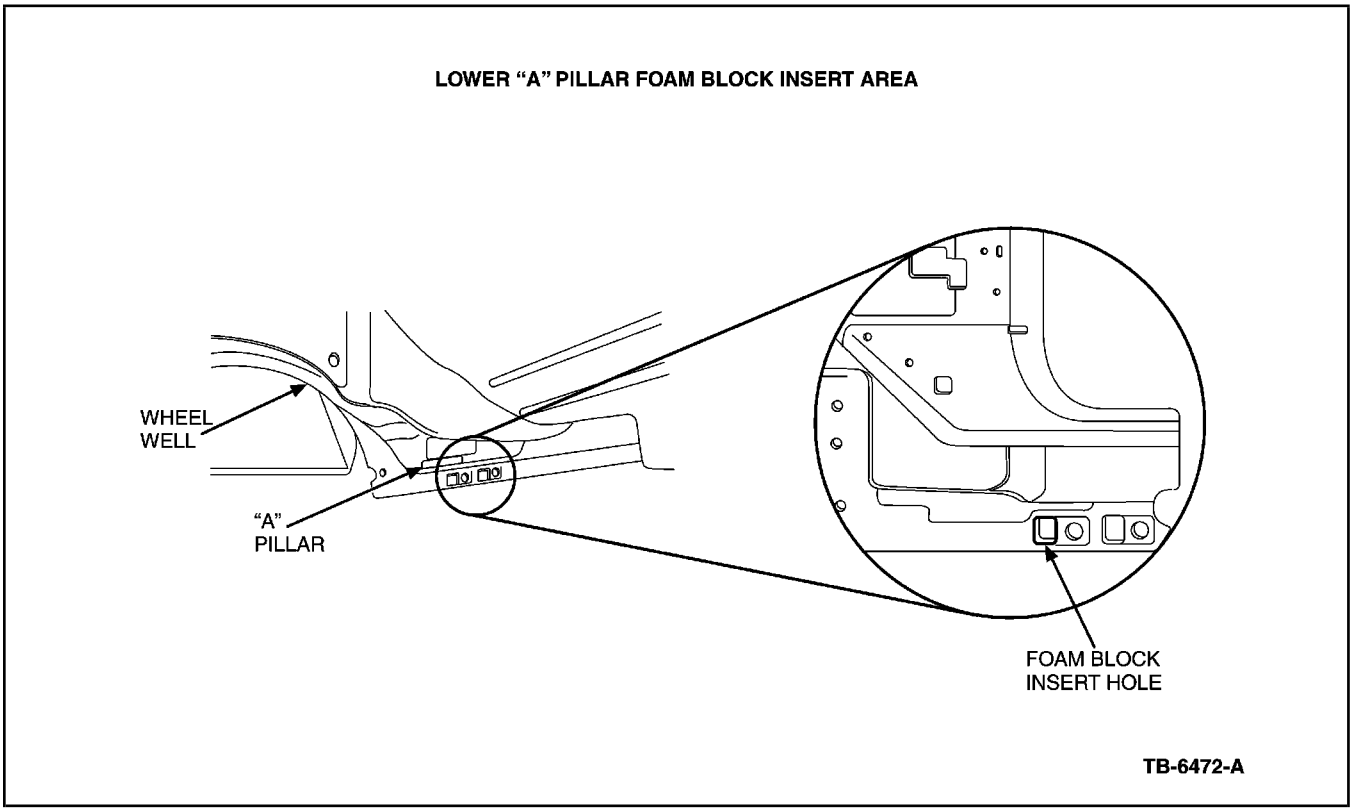


Figure 3 - Article 01-21-9

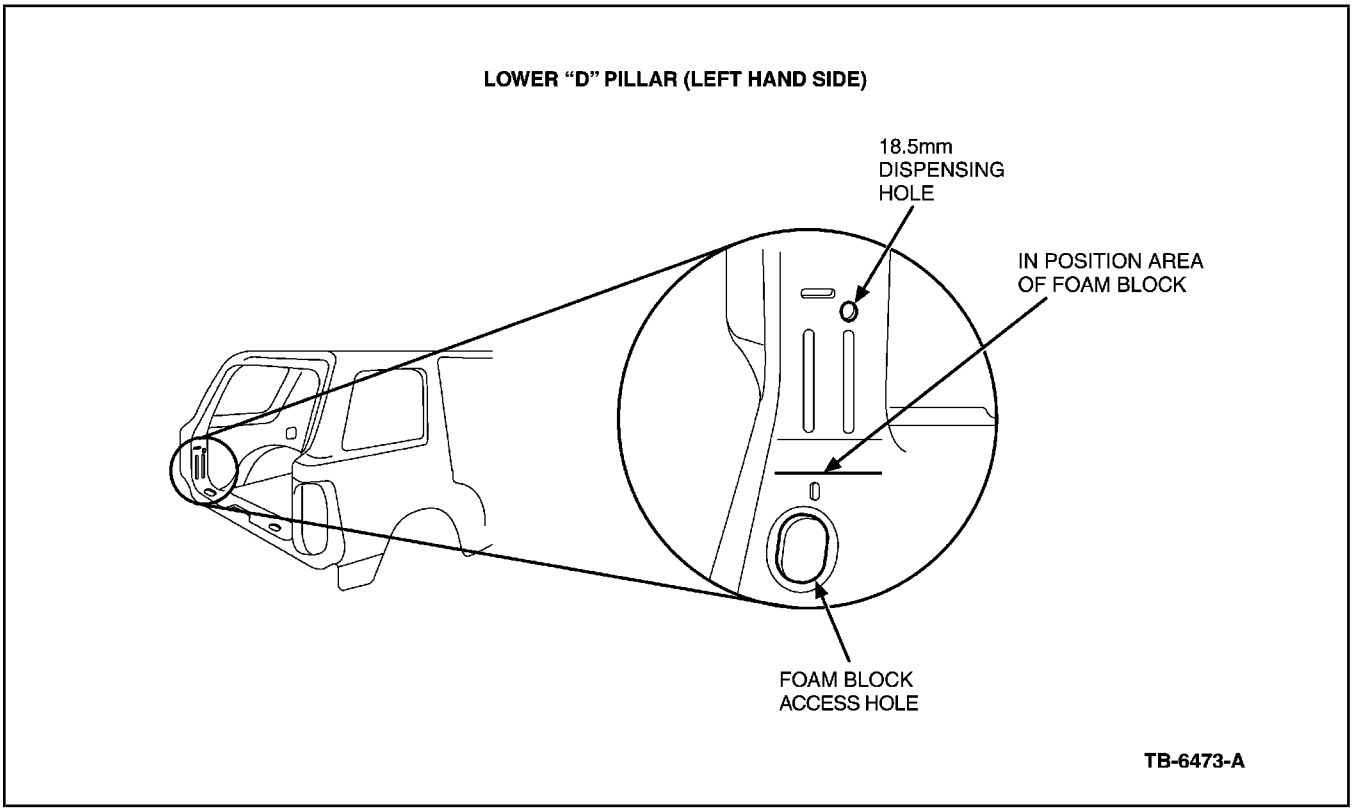


Figure 4 - Article 01-21-9

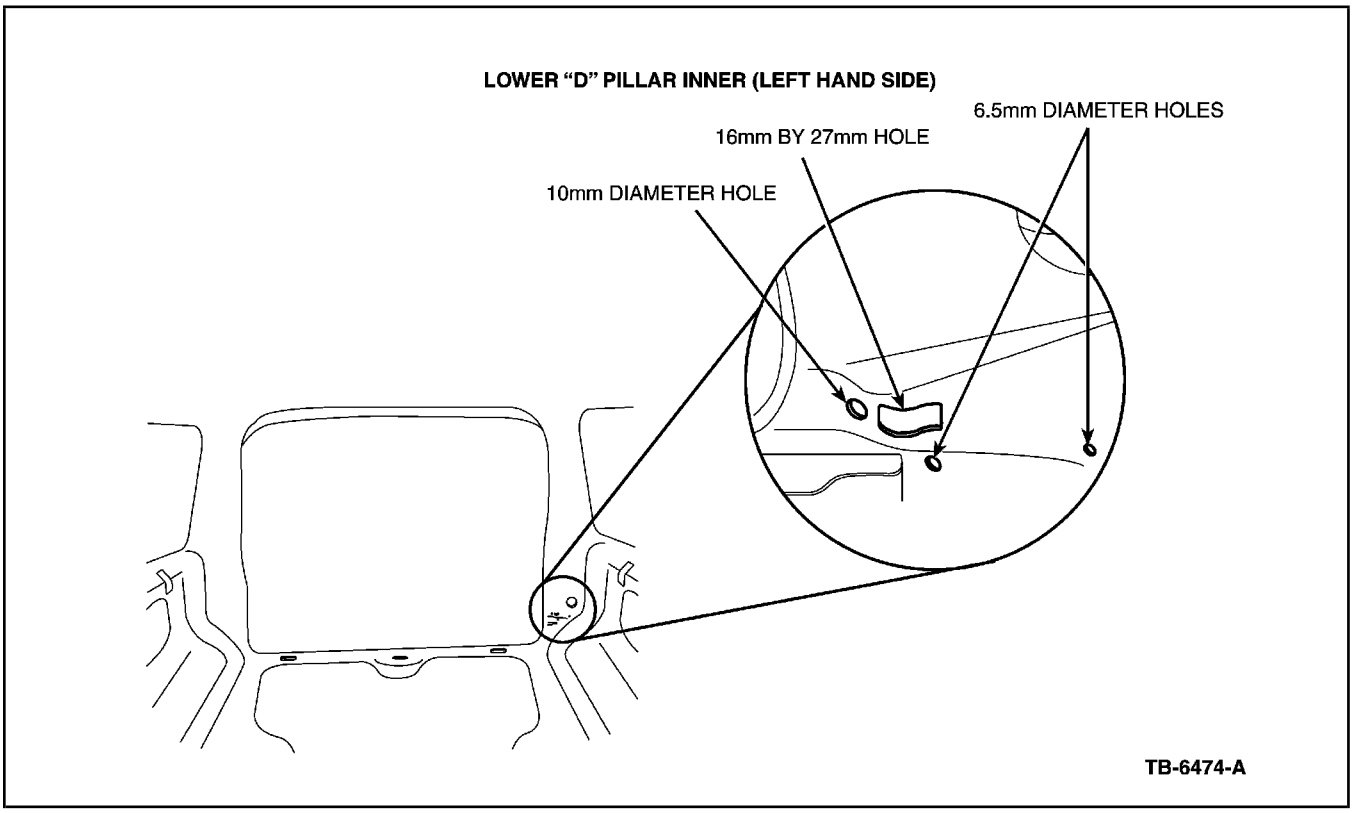


Figure 5 - Article 01-21-9

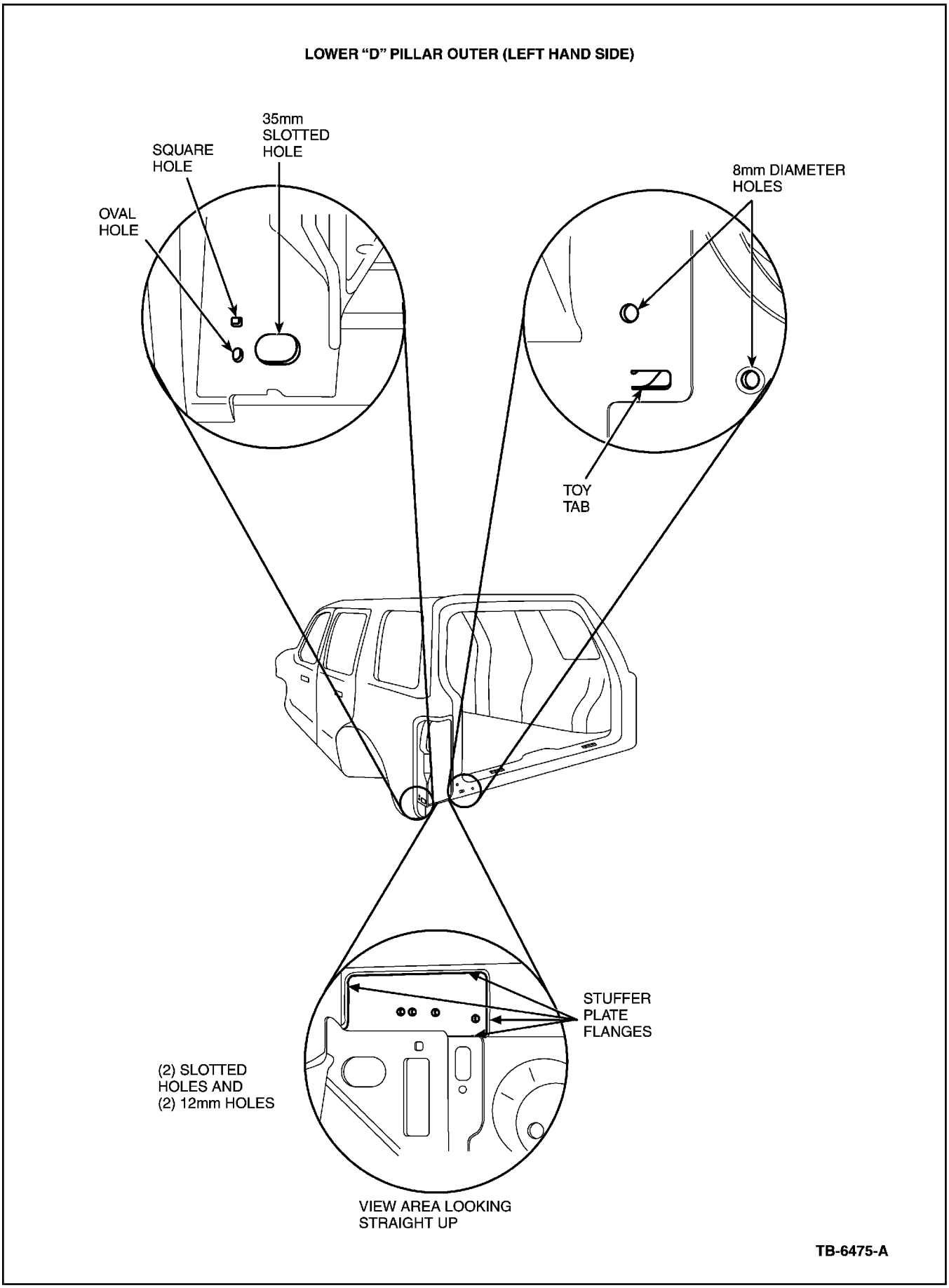


Figure 6 - Article 01-21-9

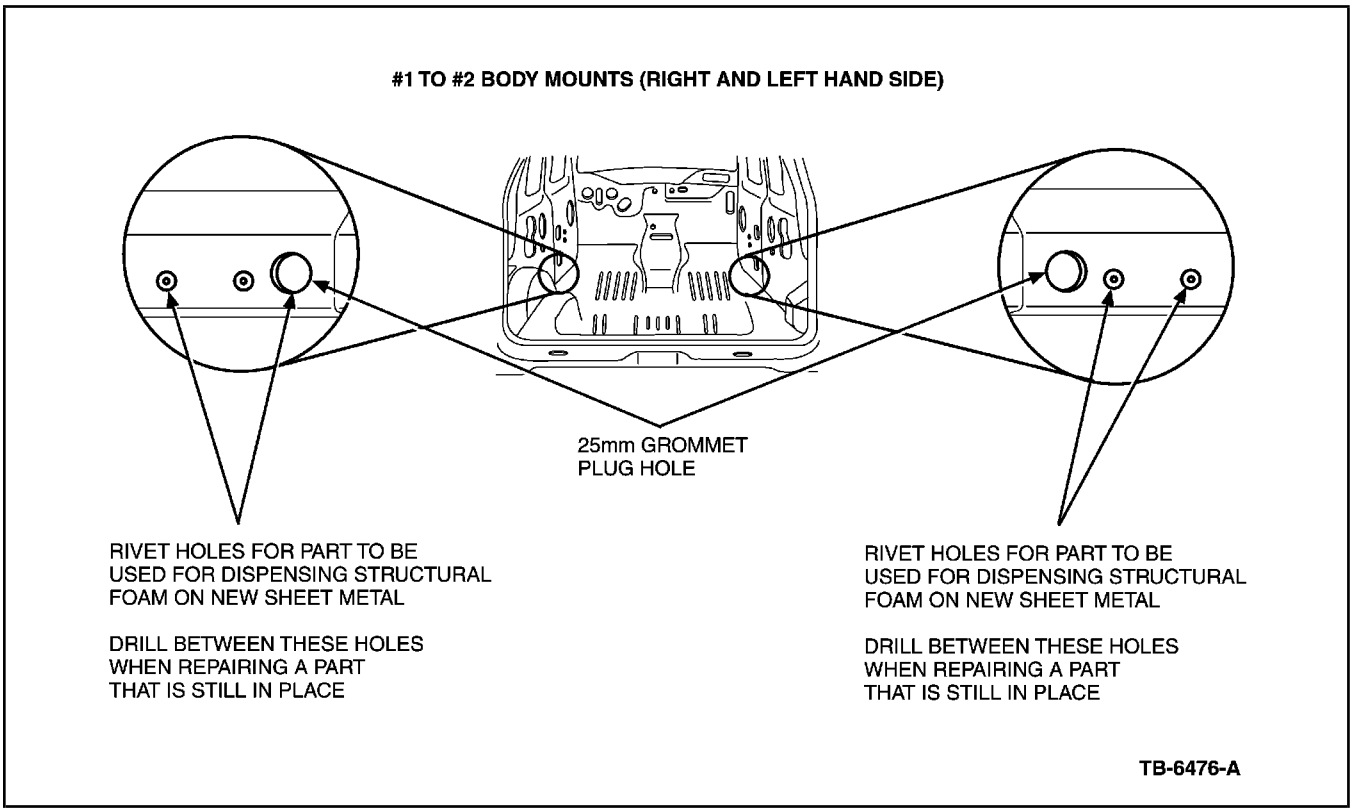


Figure 7 - Article 01-21-9

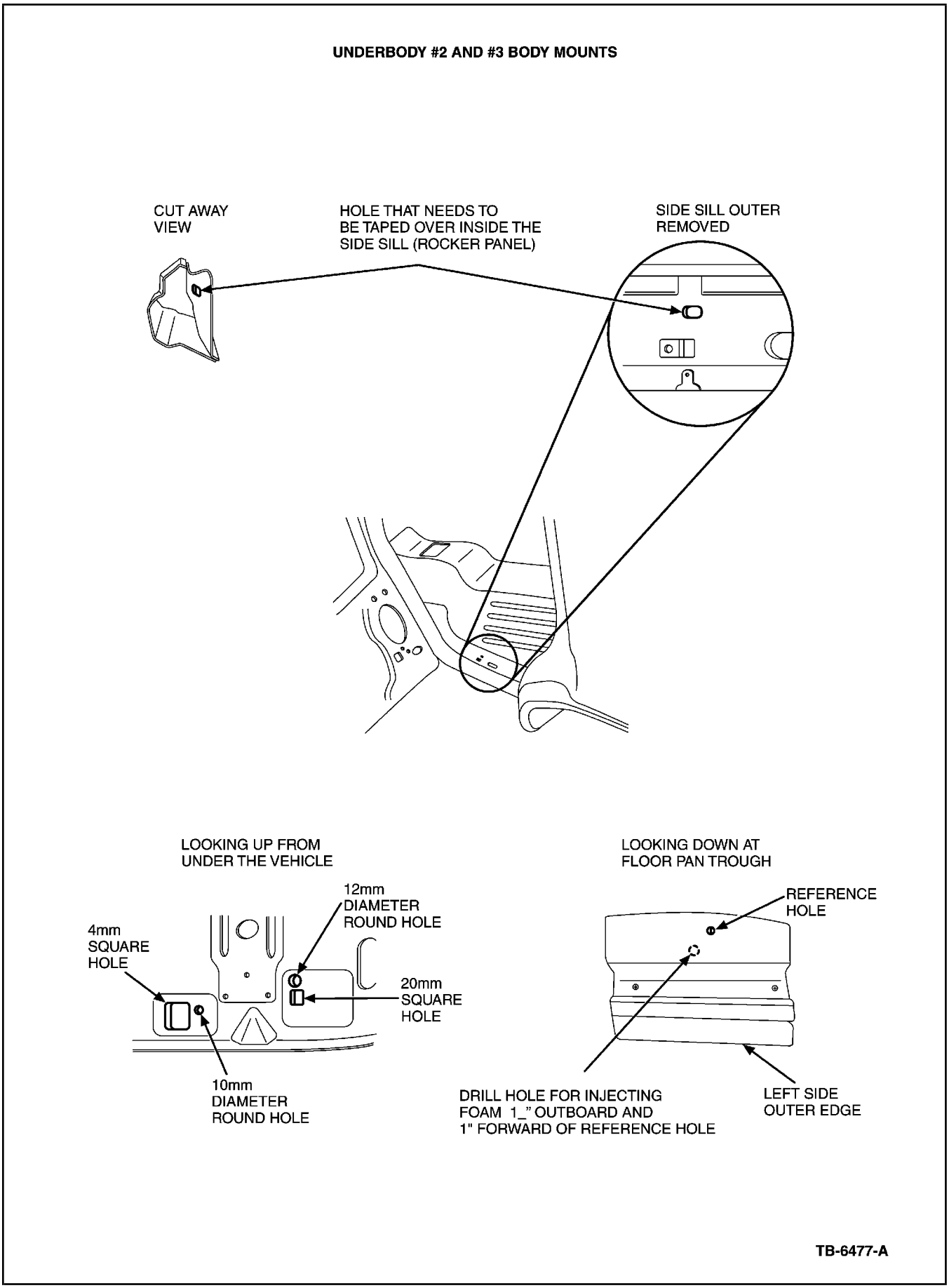


Figure 8 - Article 01-21-9

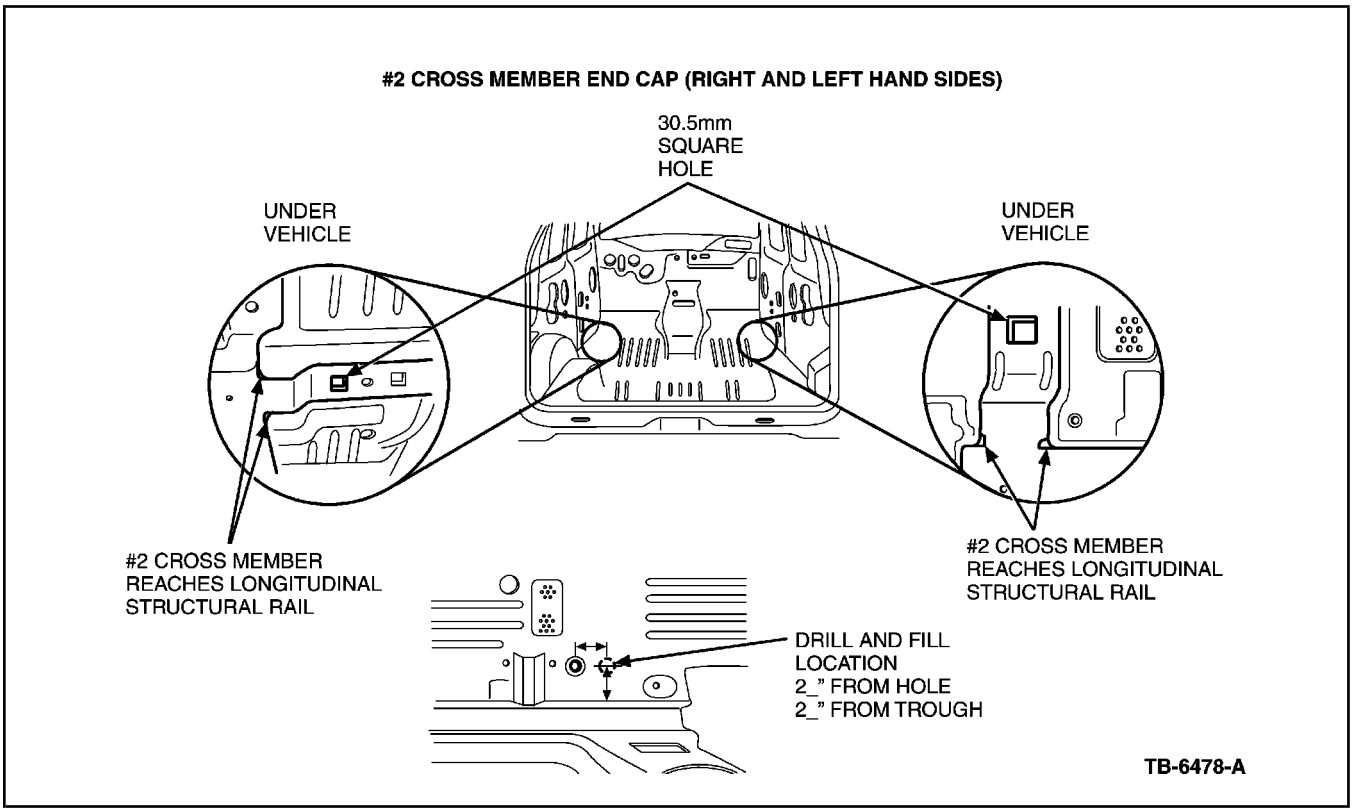


Figure 9 - Article 01-21-9

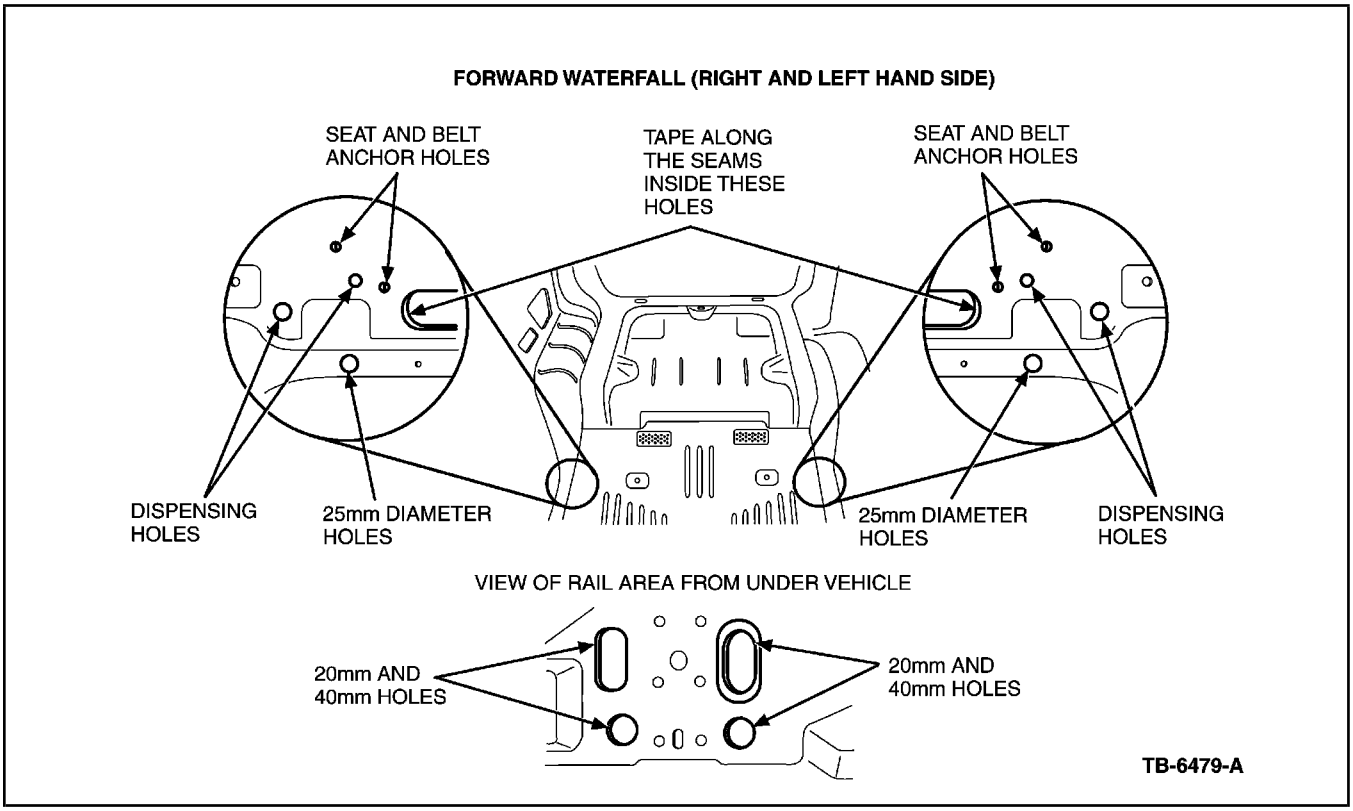


Figure 10 - Article 01-21-9

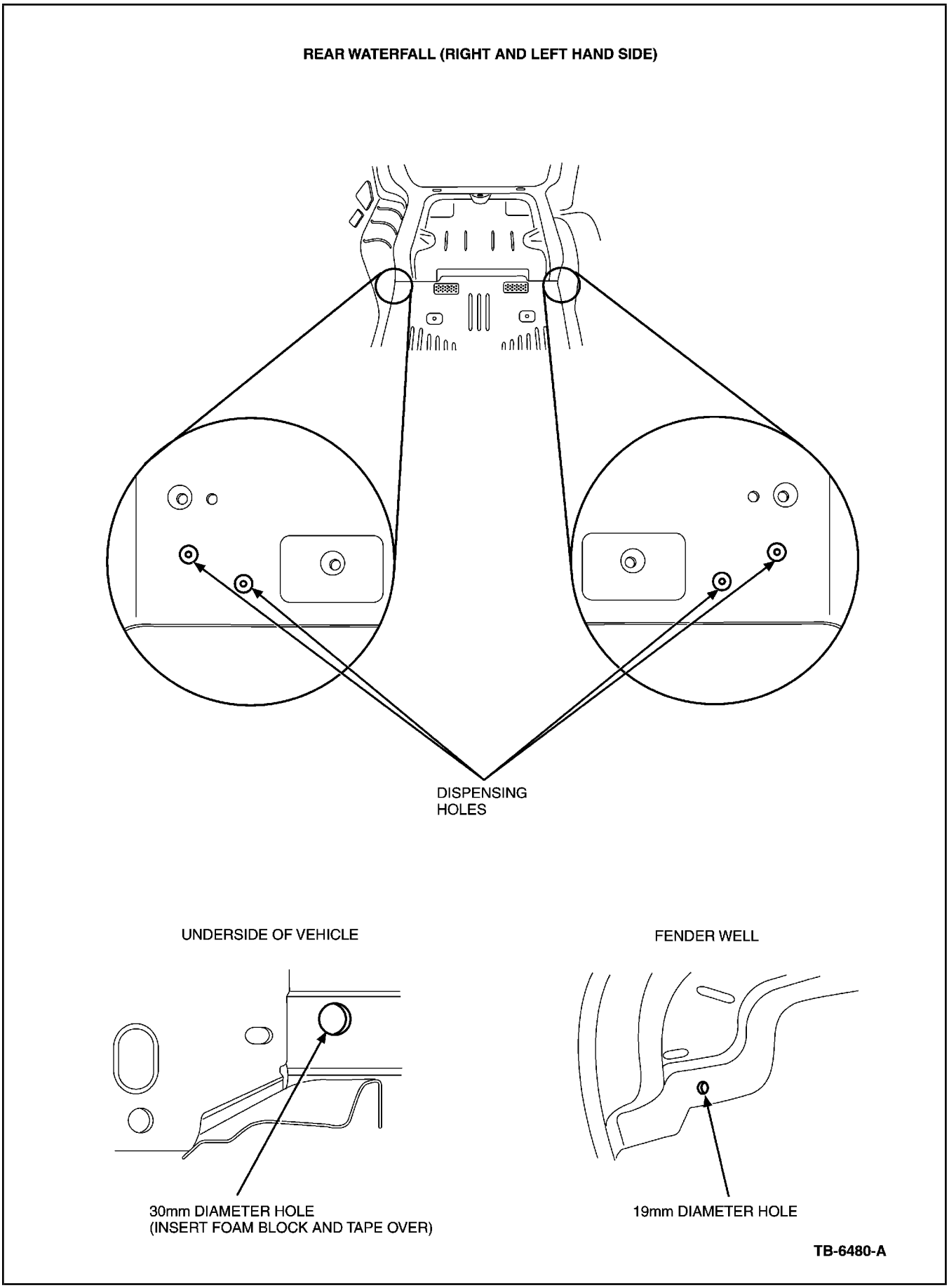


Figure 11 - Article 01-21-9